

NORTH CENTRAL WEST VIRGINIA, WV, USA

WMO: 724175

Lat: 39.296N

Lon: 80.229W

Elev: 367

StdP: 97.00

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 03802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.2	-19.3	0.7	-12.2	-17.1	0.9	-9.6	10.4	6.6	9.1	4.8	2.2	270	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	31.9	22.5	30.6	22.0	29.2	21.3	24.1	29.2	23.4	28.1	22.8	27.2	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.0	26.7	22.1	17.5	26.2	21.3	16.7	25.3	74.3	29.5	71.4	28.0	68.8	27.3	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.5	6.6	-18.1	33.2	2.9	1.6	-20.2	34.4	-21.9	35.4	-23.5	36.3	-25.6	37.5		
			-18.5	25.4	2.8	0.9	-20.5	26.0	-22.2	26.5	-23.8	27.0	-25.8	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-0.2	1.5	6.1	12.7	17.2	21.2	23.0	22.4	19.2	12.8	7.1	2.8
	DBStd	9.51	6.99	6.24	5.94	5.20	4.25	3.07	2.60	2.52	3.85	4.89	5.08	5.82
	HDD10.0	1132	324	247	152	32	3	0	0	0	0	27	114	234
	HDD18.3	2728	574	473	379	179	73	11	2	2	36	181	337	482
	CDD10.0	1937	8	8	31	112	227	336	402	386	276	114	28	10
	CDD18.3	491	0	0	1	9	39	97	144	129	61	11	0	0
	CDH23.3	4026	0	1	9	120	370	789	1124	981	528	98	3	0
	CDH26.7	1162	0	0	1	19	85	228	356	294	159	21	0	0
Wind	WSAvg	2.4	3.2	3.1	3.0	3.0	2.3	2.1	1.8	1.6	1.7	2.1	2.4	2.7
Precipitation	PrecAvg	1192	99	80	101	95	125	119	135	95	93	74	84	94
	PrecMax	1647	170	135	194	156	258	267	286	169	181	126	174	215
	PrecMin	997	43	26	38	47	48	50	72	47	23	27	22	43
	PrecStd	170	35	32	41	27	50	50	55	34	43	33	34	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	21.4	24.8	28.9	30.7	32.5	33.0	32.7	32.6	29.3	23.0	19.8
		MCWB	13.7	14.0	15.4	16.7	20.8	21.6	23.8	22.9	21.0	20.0	15.9	14.7
	2%	DB	16.4	17.3	21.9	26.7	28.8	30.8	31.4	31.2	30.3	26.0	20.7	17.2
		MCWB	12.2	11.6	13.6	16.1	19.5	21.8	23.3	22.6	21.1	18.3	13.9	12.4
	5%	DB	13.6	14.3	18.9	24.3	27.4	29.1	30.1	29.7	28.5	23.6	17.9	14.6
		MCWB	10.6	10.2	12.1	15.1	19.2	21.3	22.4	21.9	20.2	17.1	12.4	11.2
	10%	DB	10.6	11.8	16.1	22.1	25.5	27.8	28.7	28.2	26.7	21.5	16.1	12.3
		MCWB	7.9	8.1	10.9	14.0	18.3	20.8	21.8	21.4	19.4	16.1	11.7	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.3	16.4	19.0	22.6	24.0	25.3	25.0	23.7	21.8	17.1	15.5
		MCDB	18.2	19.3	23.3	25.7	28.5	29.0	30.9	30.3	29.4	26.4	21.9	18.8
	2%	WB	12.9	13.1	14.9	17.6	21.2	23.1	24.4	23.9	22.5	20.0	15.2	13.7
		MCDB	15.7	16.3	19.9	23.3	26.9	28.1	29.6	28.7	27.5	24.5	19.1	16.2
	5%	WB	11.1	10.6	13.4	16.4	20.2	22.3	23.6	23.2	21.5	18.4	13.7	11.8
		MCDB	13.3	13.7	17.5	21.9	25.3	27.4	28.3	27.6	25.9	22.2	17.0	14.0
	10%	WB	7.9	8.0	11.7	15.3	19.2	21.6	22.9	22.5	20.6	16.7	12.1	9.6
		MCDB	10.2	11.1	15.5	20.6	23.8	26.5	27.0	26.6	24.6	20.3	15.3	12.0
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.5	10.8	12.3	11.8	11.0	10.4	10.8	11.8	11.5	10.6	8.7
		MCDBR	11.7	13.8	15.0	15.7	14.3	12.7	12.0	12.7	14.3	14.4	14.4	12.7
		MCWBR	8.9	10.1	9.0	7.9	6.7	5.6	5.0	5.2	6.1	7.6	9.1	9.2
	5% WB	MCDBR	11.0	12.3	12.6	12.5	11.8	10.8	9.9	10.2	11.3	11.9	11.5	11.5
		MCWBR	9.0	10.1	8.6	7.1	6.0	5.2	4.7	4.8	5.3	6.7	8.3	9.4
Clear-Sky Solar Irradiance	taub	0.301	0.320	0.346	0.385	0.429	0.461	0.482	0.497	0.407	0.363	0.324	0.300	
	taud	2.469	2.403	2.403	2.322	2.205	2.152	2.111	2.039	2.285	2.408	2.492	2.510	
	Ebn at Noon	874	902	910	891	854	824	802	777	834	839	839	850	
	Edh at Noon	79	97	108	124	143	151	156	163	120	96	77	71	
All-Sky Solar Radiation	RadAvg	1.66	2.46	3.52	4.67	5.29	5.84	5.64	5.23	4.34	3.05	2.11	1.46	
	RadStd	0.19	0.37	0.28	0.40	0.44	0.42	0.33	0.29	0.47	0.36	0.26	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	N/A

Nomenclature: See separate page